Lnv Lochan Ch

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Dual-degree CSE and Data Science student and AI startup co-founder who builds and runs quantitative trading systems. Top 6% (302 of 5,111) in Wunder Fund's HFT order-book prediction competition, 21st in AlphaNova Cycle 4, 620th globally in IMC Prosperity, 7th of 190 in the APAC Quant Trading Hackathon, and 3rd prize in IIT Madras and NIT Rourkela's Pitch Me a Stock equity research competition. Founded my university's first Finance Club.

Education

B.S. in Data Science and Applications, Indian Institute of Technology Madras 2025 – 2029 | CGPA 7.00/10
B.Tech, Computer Science and Engineering, C. R. Rao Advanced Institute of Mathematics, Statistics and Computer Science 2025 – 2029 | CGPA 9.75/10

Quantitative Research & Competitions

Wunder Fund Alfa Connectome: HFT Order-Book Prediction, Team Varanasi Ongoing: public rank 302 of 5,111
Python, PyTorch, ONNX

- Built a stateful GRU that predicts two future-price indicators row by row from a live stream of 112 limit-order-book features on two instruments; **0.687** weighted Pearson on validation against the official baseline's 0.617.
- Met a 1-vCPU, ∼90-microsecond-per-row latency budget for 39.4M test rows: int8-quantised ONNX export (score loss under 0.0001) cut inference from 203 to **59 microseconds** per row, so a two-model ensemble runs in about 34 of the 60 minutes allowed.
- Treated every change as a controlled experiment: a baseline calibration submission showed the validation-to-leaderboard drop was shared by everyone, and seed noise (0.0013) was measured before any gain was accepted. Climbed from rank 543 to **302** with 6 submissions.

Pitch Me a Stock, IIT Madras × NIT Rourkela, Team Red Devils Oct 2026: 3rd prize; final 10 of 626
Equity research, financial modelling, valuation

- Cleared a finance quiz and a written equity research round to reach the 10-team offline final at IIT Madras from 626 participants, then won **3rd prize** presenting to the judging panel.
- Pitched a BUY on Hindustan Aeronautics (NSE: HAL) with an INR 5,520 target (+13.8%): built FY27–28 earnings forecasts from filings and management guidance, valued at HAL's 5-year median P/E, with bear, base and bull cases (−14%, +14%, +39%).
- Framed the thesis around delivery conversion (GE engine supply, aircraft handovers, revenue), checked every claim against the earnings-call transcript and Ministry of Defence releases, and set dated triggers for a downgrade.

APAC Quant Trading Hackathon (Roostoo), Solo Entry (Prime Execution) Live 9–18 Oct 2026: rank 7 of 190
Python, AWS EC2

- Built an autonomous spot-trading bot (HMAC-signed REST client, drawdown risk ladder, order planner, holdings reconciled from the exchange) that traded unattended on AWS EC2 for all 10 competition days, covered by 50 automated tests including a signature-verifying mock exchange.
- Walk-forward tested 3 signal strategies and a long/short variant on 21 out-of-sample months of Binance data against pre-declared gates (costs, 2× costs, fold stability, ±30% parameter stability, deflated Sharpe). None passed; a BTC trend filter that met every pre-set criterion was still not deployed (deflated-Sharpe probability 0.27 over 31 configurations). All 19 trials are logged.
- Live result: final equity −0.04% with a deepest drawdown of −3.14%, inside the −4% de-risk level throughout; average slippage +0.28 bps over 35 fills.

AlphaNova Biweekly Competition, Season 1 Cycle 4: rank 21 (star signal); Cycle 5 ongoing
Python, XGBoost

- Forecast a hidden cross-sectional target on 20 assets, scored by Sharpe after a 5 bp turnover cost. Built 15 candidate predictors, a walk-forward harness, an overfitting-gate simulator and automated leakage, determinism and timing checks.
- Submitted a factor-hedged reversal signal (PCA-neutralised, with an XGBoost overlay) that passed the novelty and overfitting gates; reproduced the platform's novelty metric offline (25.5° vs 25.49° on the site). A star marks a prize-eligible signal (correlation of at most 0.5 with every higher-ranked signal).
- Found that turnover decides net Sharpe: slower signals cut costs from 57–87% to 1.5–3.5% of gross PnL in local tests.
- Cycle 5: identified the hidden target as roughly a z-scored forward 120-hour return (correlation 0.93) and found two new edges, per-asset trend persistence and market lead/lag. Built a novelty model from 106 simple signals and their pairwise combinations that matched all 10 of the site's novelty verdicts.
- Cycle 5: traced an overfitting-gate failure to slow features that stay smooth under label shuffling; rebuilt the signal so it beat all 100 shuffled retrains in the hardest test months, with predictions unchanged (to 1e-14) under row padding and pickle reload.

IMC Prosperity 4 Trading Competition Global rank 620 after Round 4

Python, Rust

- Ran statistical research on the Round 3 products (stationarity and regime tests, GARCH volatility, order-flow imbalance, cointegration) and derived inventory-skewed market making, z-score mean reversion, pairs trading, and Black-Scholes option market making with delta hedging.
- Compared strategies on a Rust-backed Monte Carlo simulator (1,000 sessions in about 56 seconds), judging PnL distribution, stability and profitability rather than a single backtest.

Susquehanna Predictions Cup 2026 Rank 365 of 1,403 (12 Oct 2026)

Python

- Built an autonomous prediction-market trading system (179 commits) that scans all open election markets, ranks edges by comparing forecast probability with executable ask prices, and places limit orders.
- Added risk controls: portfolio and open-order reconciliation, idempotent submissions, per-order and aggregate exposure caps, and a dry-run rotation analyser for replacing weak positions.

Code for the private Varanasi, AlphaNova and Prime Execution repositories is available on request (Varanasi after the competition ends, as its rules require).

Research & Publications

- Co-author, "Does Pretraining Solve Temporal Domain Shift? A Multi-Tier Study of Hand-Crafted, Supervised, and Self-Supervised Representations for Music Emotion Recognition", with Prof. G. Radha Krishna; manuscript under review at ICASSP 2027.

Experience

Co-Founder, AI Solutions & Business Development, Kinetic Flow AI Oct 2025 – Present

- Co-founded an AI solutions startup that delivered 4 production-grade client solutions worth over INR 2 lakh; turned client requirements into technical specifications and handled pricing, timelines, client communication and deployments.
- Built multi-portal apps, workflow automations and backend services with Supabase (PostgreSQL), Ollama, Claude and Gemini.

Founder & Coordinator, Finance Club, C. R. Rao AIMSCS Jul 2026 – Present

- Founded the university's first Finance Club, leading vision, recruitment and operations from scratch; organized workshops on financial markets, quantitative finance and trading, and planning university-wide paper trading competitions.

Student Organizer, GDG on Campus, C. R. Rao AIMSCS 2026 – 2027

- One of three organizers of the university's Google Developer Groups chapter for the 2026–27 tenure.

Software Projects

- **DeepFake Audio Detection** (Python, PyTorch, Ollama, Node.js): PyTorch deepfake audio detector with GPU-accelerated inference; backend APIs and a Node.js web app with a local Ollama LLM.
- **GalliBite, HOMYGO, MitraCare** (Supabase, PostgreSQL, Edge Functions, WhatsApp Business): contributed to multi-portal food delivery, caregiver booking and elderly-care platforms; built auth, Edge Functions, AI support automation and real-time features. All delivered through Kinetic Flow AI.

Other Competitions & Certifications

- QuantStorm 2026 (IIT Delhi, IIT Guwahati, IIT Bombay): submitted an algorithmic trading bot for a two-player quoting and auction game; Round 1 certificate.
- IIT Bombay BodhiCoder: ranked 61 of 4,095 in the C++ program. Profit+ Trading Fundamentals, IIT Guwahati: Certificate of Merit. Ventorra Finance Olympiad: 94% in Stage 1.

Skills

- **Mathematics:** Probability, Statistics, Linear Algebra, Calculus, Optimization, Time-Series Analysis
- **Languages:** Python, C++, C, SQL, JavaScript, Rust (backtesting tools)
- **ML & AI:** PyTorch, ONNX (int8 quantisation), XGBoost, NumPy, Pandas, Matplotlib, LLMs, Ollama, OpenAI API, prompt engineering
- **Quant & trading:** walk-forward and Monte Carlo backtesting, overfitting and leakage checks, turnover and risk control, execution systems
- **Finance:** equity research, financial statement analysis, earnings forecasting, P/E-based valuation and scenario analysis
- **Backend & tools:** FastAPI, REST APIs, SQL and DBMS (Supabase / PostgreSQL), Node.js, AWS EC2, Git, GitHub, Conda
- **Training:** IIIT Hyderabad iHub AIML Training 2026; 20+ Google Cloud Skills Boost badges